IBVAP Command Center

Frontend guide (final): kya hai, andar kaise chalta hai, aur demo kaise dikhana hai · Smart India Hackathon, PS 26187

> **Ek line mein:** IBVAP border ke **pehle se lage CCTV cameras** ko AI se smart banata hai, bina naye camera ke. Ek Python AI engine (har BOP par control-room PC ya chhota AI box) video dekhta hai aur events backend ko bhejta hai; ye frontend us data ko operator ke liye Command Center ki tarah dikhata hai. **Abhi backend nahi bana hai**, isliye frontend "mock mode" mein apna nakli data khud banata hai.

1. Kaise chalayein

1. `cd frontend` → `npm install` (sirf pehli baar)
2. `npm run dev` → browser mein `http://localhost:5173`
3. Login par role chuno (Operator / Commander / Admin) → **Enter Command Center**. Password nahi chahiye (mock).
4. Demo ke liye: andar jaakar `Ctrl+Shift+D` ya Settings → Demo Mode.
5. Production build: `npm run build` phir `npm run preview` (port 4173).

`.env` mein `VITE_USE_MOCK=true` = nakli data. Jab backend ban jaye: `false` karke `VITE_API_URL` / `VITE_WS_URL` do.

2. Andar kaise chalta hai

```
Pages → Zustand store (data / ui / auth) → [redacted] (DataService) → mockService.ts  YA
                                                                      apiService.ts (REST + WebSocket)
```

- **Service layer:** koi page seedha `fetch` nahi karta. Sab `service` se. Isliye mock se real backend par switch sirf ek setting hai.
- **Ek live subscription:** login ke baad store `start()` hota hai: data load, alert stream subscribe, toasts / sound / dialog sab yahin se.
- **Mock simulator:** har ~40 s ek random alert (`MOCK_ALERT_INTERVAL_MS`), har 4-9 s ek plate read, camera boxes 10 baar/second. Memory capped hai (alerts 800, plate reads 600), lambe demo mein slow nahi hota.
- **Incidents:** ek hi object ke baar-baar aaye alerts ek incident ban jaate hain (same camera + track 10 min tak, ya same camera + type 2 min tak). Card par `x3` jaisa counter.
- **Offline:** link toote to cached data dikhta hai, operator ke actions queue mein rehte hain aur link aate hi sync hote hain.
- **Contract:** `frontend/docs/API_CONTRACT.md` mein har endpoint, JSON shape aur AI-engine ingest likha hai. Backend wala isi se banayega.

3. Notification policy (alarm-management standard)

Severity	Kya hota hai
CRITICAL	Beep + animated toast + bell pulse. Fence intrusion / face match par full-screen dialog jisme **clip loop** chalta hai.
HIGH	Chupchap toast, koi awaaz nahi.
MEDIUM / LOW	Koi toast nahi, sirf live feed mein.

- Notification **ek incident par ek baar**. Repeat alert sirf counter badhata hai.
- CRITICAL acknowledge na ho to **60 s par ek reminder**, **3 min par apne aap "Escalated to supervisor"** (history + audit). Baar-baar beep nahi.
- **Low likelihood** alert kabhi beep nahi karta. Shelved incident aur out-of-service camera notify nahi karte.

4. Jab ek alert aata hai

```
AI engine → backend → WebSocket → store.ingest() → incident check → policy → toast / beep /
                        dialog → camera tile flash → map marker pulse
```

Card par dikhta hai: type, subtype, likelihood (High/Medium/Low, text + icon), "why it fired" (frames, confidence), AI degraded (fog), drone watch tag. Operator Acknowledge / Escalate / False alarm / Shelve karta hai; har action history aur audit log mein jaata hai.

5. Roles

Kaam	Operator	Commander	Admin
Acknowledge, Escalate, False alarm, Shelve, On hold, Owner assign, drone watch, share with naka, notify QRT	**Haan**	**Haan**	**Haan**
Resolve (SOP ke zaroori steps pure hone par hi)	**Nahi**	**Haan**	**Haan**
Unshelve pehle, camera out of service, zones / watchlist / blacklist / settings edit, Audit Log page	**Nahi**	**Haan**	**Haan**

6. Har page kya karta hai

Command Center

- 4 KPI tiles (normal = grey, problem = rang)
- Sector overview: har BOP ke khule incidents, last shift handover
- 4 live feeds, 60 min timeline, live alert feed

Live Cameras

- 1x1 / 2x2 / 3x3 (max 9 feeds), Sector → BOP → Camera tree se filter
- Thermal camera thermal rang mein, IR night tag, "AI degraded (fog)"
- Out of service (Commander), snapshot

Map View

- Cameras, nakas, indicative border line, heat circles
- **Mark drone sighting:** map par click, 1/2/5 km, 2/4/6 h → watch circle, andar ke cameras / ANPR / nakas highlight, top bar countdown

Alerts + Alert Detail

- Incidents / All / Shelved tabs, filters (subtype bhi), CSV
- **Clip player:** -10 s se +20 s, alert moment mark, 0.5x/1x/2x, Bookmark
- Tracking view, playbook, Notify QRT, Owner, On hold, SOP gate, evidence ZIP (SHA-256)

ANPR + Search

- ANPR: har plate read, class filter, blacklist, galat plate correct
- **Search:** person / vehicle, class, rang, disha (border ki taraf / door), camera, zone, time, plate ka hissa → thumbnails, click par clip

Face Watchlist + Zones

- Watchlist, match ko Confirm / Reject (decision support)
- Zones: polygon + tripwire (disha ke saath), severity, classes, dwell, change log

Shift Handover

- Jaane wala: khule incidents, shelved, drone watches, band cameras + notes
- Aane wala: review tick karke **Accept handover** (audit mein)

/qrt (phone view)

- QRT ko diye incidents, bade buttons: Accept → On scene → Cleared
- Dashboard par turant dikhta hai. Notify QRT ke baad "Open QRT view".

Analytics, Reports, Audit, Edge Nodes

- Charts (type, subtype, vehicles by class, day/night, heatmap), shift report
- Audit log: kisne kya kab kiya
- Edge Nodes: har BOP ka PC / AI box, CPU / GPU / temp / FPS

Settings + About

- Cameras (type, BOP), detection, notification policy, C2 integration (REST / WS / webhook), users, privacy, security, demo
- About: PS 26187 ki har requirement kahan dikhti hai

7. Keyboard shortcuts

Key	Kaam
`A`	Sirf khule critical dialog wala ya click kiya hua alert acknowledge (mouse ke neeche wala nahi). "Acknowledged: ..." toast aata hai.
`M`	Awaaz mute / unmute
`1` `2` `3`	Camera layout 1x1 / 2x2 / 3x3
`?` / `Esc`	Shortcuts list / dialog band
`Ctrl+Shift+D`	Demo Mode start / stop

8. Demo Mode: "Punjab sector, foggy night, drone drop" (~2.5 min)

1. Raat, dhund: normal cameras par "AI degraded (fog)", thermal River Cam saaf dekhta hai (3x3 grid).
2. BOP Ajnala River Cam ke paas **DRONE_SIGHTING** alert.
3. Operator map par drone mark karta hai → **2 km drone watch**, nakas highlight, top bar countdown.
4. ICP Attari par **face watchlist match** (HIGH, chupchap toast).
5. Thermal camera: aadmi fence ke paas **rengta** hai (night movement + crawling), "In drone watch area", severity ek level upar.
6. Wahi track ruk kar ghoomta hai (loitering): incident counter badhta hai, extra beep nahi.
7. Wo Riverbed zone cross karta hai: **CRITICAL dialog clip ke saath**, operator dekh kar acknowledge karta hai.
8. Ek **jeep** nikalti hai: ANPR Road Gate par **blacklisted plate** padhta hai.
9. **Share with police naka** (simulated).
10. Playbook tick + **Notify QRT** → **/qrt** par team **Accept** karti hai.
11. Clip mein bookmark + **evidence ZIP** download (SHA-256, chain of custody).
12. Incident **escalate** → **Audit log** → **C2 integration** (REST / WS / webhook) → About page requirement table.

Demo do baar lagatar bina reload chal sakta hai.

9. Abhi kya asli nahi hai

- **Backend aur AI engine nahi bane.** Saare alerts, cameras, plates, chehre, nakas mock data hain. `apiService.ts` kabhi real server se nahi chala.
- **Video nakli hai:** camera feed aur clip canvas par bane hain ("SIMULATED FEED" / "SIMULATED CLIP" likha hai). Evidence ZIP mein clip file nahi, sirf clip window + bookmarks.
- SMS / radio relay, naka share, webhook test: kuch bhi bahar nahi jaata (simulated).
- Border line andaaze ki hai ("Indicative ... not official"). Tracking path aur search history bhi mock banata hai.
- Face blur aur data retention sirf settings hain, asli mein apply nahi hote.
- Shift handover mein aane wala aur jaane wala ek hi login ho sakta hai. Reload ke baad ek reminder dobara aa sakta hai.
- Page reload karne par mock data reset ho jaata hai.